4DV650/4DT901

Systems Modeling and Simulation - Assignment 2

In this assignment, you should work in a **Group of 2 members**. The names of both group members must be reported with the submission.

For any questions about the Assignment, use the **"Zulip Channel"** to get in contact with us.

Grading

Your submission will receive an A-F (according to ECTS grading scale). You are allowed to improve your work after the initial submission before the deadline. **The grade is final**, *i.e., you will not get an opportunity to correct/improve after grading*.

Please note that all tasks **must** be documented; that is, a figure/table without text that explains it has little or no value. In solving the tasks, you are **only allowed to use knowledge from the course material**, and whenever you do so, you **must** explicitly provide a reference to the source (indicating Chapter, Section, and page number).

Your answers should be your own! Each submission will pass through a plagiarism/clone detection system before correction. If plagiarism is detected, the assignment has failed, and a formal investigation will be initiated.

Any use of AI tools must be reported. Whenever you use an AI (e.g., ChatGPT, Copilot, Claude, …) in completing the tasks, specify **which parts were assisted** and **how** they contributed.

Evaluation Process

1. Submit a PDF, containing the report (filename: `Group_number_A1.PDF`)
2. Submit a ZIP, containing
   - Source code of your exercises (must be fully documented)
   - Binaries to be executed
   - Instructions for compiling and executing (include libraries, if needed)
   - Files containing all the raw data generated/collected

DON'T INCLUDE the PDF in the ZIP File

NOTE: All group members must click the **submit button** for the group submission to be considered as **submitted!!**

Problem Scenario

An automated fulfillment center operates a fleet of Autonomous Mobile Robots (AMRs). At the end of operational cycles, AMRs head to a centralized Robotic Maintenance Hub. Due to strict floor-space constraints, each workstation in the hub has a limited staging buffer. Arriving AMRs that cannot be accommodated in the staging buffer are redirected to an off-site secondary depot, resulting in transport delays, fleet unavailability, and operational disruption.

System Architecture & Service Workflow

All AMRs entering the hub follow a multi-stage service workflow across three specialized workstations (S1 → S2 → [S3]):

1. Stage 1 (S1) – Automated Inspection & Decontamination:
   - Every arriving AMR must pass through S1 first.
   - Service Time: Constant at 1.0 minute.
   - Capacity: 1 active station + 3 buffer spaces.
   - Redirection Rule: If S1's buffer is full upon arrival, the AMR is immediately redirected to the off-site depot.
2. Stage 2 (S2) – Main Diagnostic & Light Recalibration Cell:
   - Upon completing S1, the AMR proceeds to S2\.
   - Capacity: 1 active station + 3 buffer spaces
   - Service & Routing Classification:
     – Recalibration (45% of traffic): Service time follows an Exponential distribution with a mean of 2.0 minutes. Upon completion, the AMR exits the hub.
     – Sensor Alignment (45% of traffic): Service time follows an Erlang-2 distribution with a mean of 4.0 minutes. Upon completion, the AMR exits the hub.
     – Heavy Drive Repair (10% of traffic): Diagnostic check requires a constant 1.0 minute, after which the robot is routed to Stage 3 (S3).
3. Stage 3 (S3) – Specialized Heavy Drive Repair Cell:
   - Receives only AMRs routed from S2 requiring Heavy Drive Repair.
   - Capacity: 1 active station + 3 buffer spaces.
   - Service Time: Follows a Continuous Uniform distribution U(6.0 min, 10.0 min).
   - Redirection Rule: If S3's buffer is full when an AMR is transferred from S2, the AMR is redirected to the off-site depot.

Operational Workload

AMRs arrive at S1 according to a Poisson process at a rate of 45 AMRs per hour.

Evaluations Required

Evaluate the performance of the system and report:

1. The rate at which AMRs are redirected to the off-site depot.
2. The system performance in terms of: (a) Workstation utilization and overall system throughput. (b) Average waiting times across stages and in the overall system.
3. Carefully state any assumptions or approximations made in your model.

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Task 1: Simulation

Construct the computer program in a general-purpose language (e.g., C, Java, Phyton)

1. Describe how you plan to solve the problem
2. Specify and explain the Event-Graph
3. Describe the *Simulation Components* and *Organization* of your simulation model according to the *Next-event Time-advance Approach*. In particular:
   - How do you represent the *system state*?
   - How do you process the *events* defined by the Event-Graph?
   - What *statistics* do you need, and how do you compute them?

Task 2: Experiments

Design the set of experiments to run:

1. What type of simulation (*terminating*, *steady state*)? And why?
2. How do you design the experiments to obtain 95% confidence?
3. Collect data (raw data should be included in the ZIP archive)

Task 3: Output Analysis

1. For each measure of interest, calculate:
   - Sample Average
   - Sample Variance
   - Confidence Interval half-width
2. Validate the output obtained from the simulation

Submission Guidelines & Format Requirements

– **Page Limit**: Max 5 pages, Times New Roman pt 12. (excluding title page/references).

– **Code Raw Data**: Full simulation source code, execution scripts, and raw trace data must be submitted in the accompanying .zip archive (do not paste raw code or lengthy data logs into the report body).

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